

Knoema Engine

Open runtime for persistent NPCs, synthetic replay research, and reproducible agent simulation.

Triple-use narrative

Pedagogical criminology: fictional, non-identifying replay overlays for classroom safety discussions.
Reproducibility research: seed-fixed configs, logs, and benchmark artifacts for auditable agent studies.
Game-studio demo: persistent-memory NPC runtime with adapters, replay viewer, and voice playback surface.

Top benchmark numbers

LoCoMo-inspired long-term conversational retention proxy: 1.000 / target 0.800 (source: benchmarks/memory_benchmark_integration/results/summary.json rows[0].score,target)
MemoryAgentBench-inspired EventQA + FactConsolidation proxy: 1.000 / target 0.800 (source: benchmarks/memory_benchmark_integration/results/summary.json rows[1].score,target)
MemoryArena-inspired decision-relevant memory proxy: 1.000 / target 0.600 (source: benchmarks/memory_benchmark_integration/results/summary.json rows[2].score,target)

1K-agent scale proof

1000 agents x 100 ticks ran deterministically in 13.085s at 7642.50 agent-ticks/sec, peak RSS 144.426 MB (source: benchmarks/city_scale_1k_report.md lines 4, 5, 12-14).

Demo and paper status

HF Space: <https://huggingface.co/spaces/celovin/knoema-playground>
arXiv:<pending>



HF Space demo